Mock exam 2

Planetary Systems final exam practice • closed book • 120 minutes

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This mock exam has the same shape and level as the final exam: 6 questions of 10 points each, 60 points in total, in a 120-minute closed-book sitting where a calculator and a pen are all that is allowed. Your grade is $1 + 0.15 \times$points, rounded to the nearest half grade; the pass mark is 30 points. The twelve relations on the exam formula list are assumed known; every other formula a question needs is printed in the question itself. Marks per part are shown in brackets, and the steps carry marks, not only the number: show your algebra. Some parts state an intermediate result ("show that ..."); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps and round only at the end.

Constants and data

$G = 6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ $1\ \mathrm{AU} = 1.496 \times 10^{11}$ m $1\ \mathrm{yr} = 3.156 \times 10^{7}$ s $= 365.25$ d
$1\ \mathrm{d} = 86\,400$ s $M_\odot = 1.989 \times 10^{30}$ kg $k_B = 1.381 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ $m_u = 1.661 \times 10^{-27}$ kg
$\sigma = 5.670 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$ $S_0 = 1361\ \mathrm{W\,m^{-2}}$ (solar constant at 1 AU)

Body	a, A	Further data
Venus	0.723 AU	
Earth	1.000 AU	$R = 6371$ km, $\bar{\rho} = 5514\ \mathrm{kg\,m^{-3}}$
Mars	1.524 AU	$M = 6.417 \times 10^{23}$ kg, $R = 3390$ km
Saturn	9.58 AU, $A = 0.34$	$M = 5.683 \times 10^{26}$ kg, $R = 58\,232$ km, effective temperature 95 K, outer edge of the main rings at 136 800 km from the centre
Titan		$M = 1.345 \times 10^{23}$ kg, $R = 2575$ km, upper-atmosphere $T = 160$ K
Moon		$M = 7.342 \times 10^{22}$ kg, $R = 1737$ km, dayside surface $T = 390$ K
Comet	$q = 0.59$ AU, $Q = 35.1$ AU	

A total power L spread over a sphere of radius r gives a flux $L/(4\pi r^2)$; for sunlight this means $S = S_0/a^2$ with a in AU.

Problem 1 The orbit of a comet

A newly discovered comet orbits the Sun with perihelion distance q and aphelion distance Q as listed in the data table. For an ellipse with semi-major axis a and eccentricity e, the perihelion and aphelion distances are

$$q = a(1 - e), \qquad Q = a(1 + e).$$

The vis-viva equation gives the speed at distance r on an orbit with semi-major axis a:

$$v^2 = GM_\odot\left(\frac{2}{r} - \frac{1}{a}\right),$$

and at the two extremes of the orbit, where the velocity is perpendicular to the radius, conservation of angular momentum gives $v_\mathrm{p}\, q = v_\mathrm{a}\, Q$.

(a) **[3 pts]** Compute the semi-major axis, the eccentricity, and the orbital period of the comet.

(b) **[4 pts]** Show that the comet's speed at perihelion is 54.4 km s^{-1}, then compute its speed at aphelion.

(c) **[3 pts]** Judge this claim in 2 to 3 sentences, using your result from part (b): *"A comet on this orbit spends most of its time in the inner solar system, where it is bright and easy to observe."*

Problem 2 The energy of building Mars

Consider the assembly of Mars from planetesimals. The gravitational binding energy of a uniform sphere of mass M and radius R,

$$E = \frac{3}{5}\frac{GM^2}{R},$$

is the energy released when the body is assembled from material dispersed to infinity. The heat needed to warm a mass M by ΔT is $E = Mc_p\Delta T$; for rock, $c_p \approx 1000$ J kg^{-1} K^{-1}, and silicate rock starts to melt near 1400 K.

(a) **[3 pts]** Show that the accretion of Mars released about 4.9×10^{30} J.

(b) **[3 pts]** Assume for the moment that all of this energy is retained as heat. Compute the temperature rise of Mars and compare it with the melting temperature of rock. Explain in 1 to 2 sentences why the actual temperature rise was smaller, and what the comparison nevertheless implies for the early state of Mars.

(c) **[4 pts]** Judge this claim in 3 to 4 sentences: *"A planet without a global magnetic field today cannot have an iron core."*

Problem 3 Inside the Earth

Earth's radius and mean density are listed in the data table. For this problem, model the interior as two uniform layers: an iron core of density ρ_c = 11 000 kg m^{-3} and a rocky mantle of density ρ_m = 4500 kg m^{-3} (both values already include compression). The mean density of such a body is the volume-weighted average of the two layer densities,

$$\bar{\rho} = \rho_\mathrm{c}\, x^3 + \rho_\mathrm{m}\left(1 - x^3\right), \qquad x = \frac{R_\mathrm{c}}{R},$$

where R_c is the core radius.

(a) **[4 pts]** Show that the two-layer model puts the core radius near 3400 km, then compute the fraction of Earth's mass that resides in the core.

(b) **[3 pts]** Inside a uniform sphere of density ρ, the local gravitational acceleration is

$$g(r) = \frac{4}{3}\pi G\rho\, r.$$

Integrate the equation of hydrostatic equilibrium from the surface to the centre to show that the central pressure of a uniform sphere is

$$P_c = \frac{2}{3}\pi G\bar{\rho}^2 R^2,$$

and evaluate it for Earth. The measured central pressure is 364 GPa; explain in 1 sentence why the uniform estimate falls short.

(c) **[3 pts]** Judge this claim in 2 to 3 sentences: *"Earth's magnetic field is generated in the solid inner core."*

Problem 4 The rings and glow of Saturn

Saturn's properties are listed in the data table. A strengthless (fluid) satellite of density ρ_m is torn apart by tides inside the Roche limit

$$d = 2.44\,R_p\left(\frac{\rho_p}{\rho_m}\right)^{1/3},$$

where R_p and ρ_p are the radius and mean density of the planet.

(a) **[3 pts]** Compute Saturn's mean density and compare it with the density of liquid water.

(b) **[4 pts]** The ring particles are water ice with $\rho_m = 900\ \mathrm{kg\,m^{-3}}$. Show that the Roche limit lies near 130 000 km from Saturn's centre. Compare it with the outer edge of the main rings and with the fact that all of Saturn's sizeable moons orbit beyond 180 000 km, and state what this comparison says about why the rings exist.

(c) **[3 pts]** Compute the sunlight flux Saturn absorbs per square metre of its surface. Then use the measured effective temperature (data table) to show that Saturn emits about 1.9 times the power it absorbs, and explain in 1 to 2 sentences where the excess comes from.

Problem 5 Keeping an atmosphere

Whether a body keeps an atmosphere depends on the race between its gravity and the thermal motion of gas molecules. A molecule of mass m in a gas at temperature T has the most probable speed

$$v_{th} = \sqrt{\frac{2k_B T}{m}},$$

and a useful rule of thumb is that a gas survives for billions of years only if the escape velocity exceeds about $6\,v_{th}$ at the temperature of the upper atmosphere. Molecular masses: $m = \mu\, m_u$ with $\mu = 28$ for N_2 and $\mu = 2$ for H_2.

(a) **[3 pts]** Compute the escape velocity from Titan's surface.

(b) **[4 pts]** Compute the most probable thermal speeds of N_2 and H_2 at the temperature of Titan's upper atmosphere, and apply the rule of thumb to each gas. What does the result explain about Titan's atmosphere?

(c) **[3 pts]** The Moon's escape velocity is close to Titan's, yet the Moon has no atmosphere. Compute the numbers that resolve this apparent contradiction.

Problem 6 The habitable zone

The habitable zone is the range of orbital distances in which a rocky planet with an Earth-like atmosphere can sustain liquid water on its surface. Climate models place it where the stellar flux S lies between about $0.35\,S_0$ (outer edge, maximum greenhouse) and $1.05\,S_0$ (inner edge, runaway greenhouse), with S_0 the flux Earth receives. For a star of luminosity L, the flux at distance a (in AU) is

$$S = \frac{L}{L_\odot}\frac{S_0}{a^2}.$$

(a) **[3 pts]** Compute the inner and outer edge of the Sun's habitable zone, in AU. Which of the planets of the solar system orbit inside it?

(b) **[4 pts]** A red dwarf star has $L = 0.02\,L_\odot$ and $M_* = 0.30\,M_\odot$. Compute the edges of its habitable zone, and the orbital period of a planet at the inner edge.

(c) **[3 pts]** Judge this claim in 2 to 3 sentences, using your result from part (a): *"Mars orbits inside the Sun's habitable zone, so Mars is habitable today."*